

VI SEM SYLLABUS - 2026**1. Electronic Content Design****Module -I**

Introduction to E-learning- Definition, history, benefits and drawbacks of online learning, best practices of online learning, future of e learning. Overview of LMS. Technologies used in e - learning, Online course, tools to create an online course, need of the millennial learners, 21st century skills and E-learning trends.

Module -II

E-content. Designing and Development of E-content. Standards of E-content. Learning Objects and Re-usability of E- content. Phases of e-content development, various instructional models- ADDIE and ASSURE instructional model. An overview of Content Authoring Tools.

Module -III

Principles of Visual Design- Visual hierarchy, typography, and colour theory, User Experience (UX) Design- UX principles, wireframing, and prototyping, Graphic Design for Digital Media- Image editing, resolution, and file formats.

Module -IV

HTML and CSS basics, designing for different screen sizes and devices, Multimedia Content Creation- Creating multimedia elements (images, audio, video), Social media platforms and content planning, Develop a social media content calendar, Content Management Systems (CMS), introduction to CMS platforms (e.g., WordPress), search Engine Optimization (SEO), SEO basics and best practices, Copyright and Ethics in Digital Content, Copyright laws, fair use, and ethical considerations, Content Evaluation and Feedback, Usability testing and feedback collection.

Textbooks

1. Diane Elkins et al , E-Learning Fundamentals: A PRACTICAL GUIDE, . (2015). ISBN: 9781562869472, Pages: 176.

Reference books

1. "The Non-Designer's Design Book" by Robin Williams.
2. "E-Learning and the Science of Instruction" by Ruth C. Clark and Richard E. Mayer.
3. "HTML and CSS: Design and Build Websites" by Jon Duckett.

2. SOFTWARE TESTING**Module -I****(12 Hours)**

Introduction: Basic definitions, A testing life cycle, Test Cases, Fundamental approaches to apply Test Cases, Levels of Testing, Examples: The NextDate function, Triangle problem and The Commission Problem and The SATM (Simple Automatic Teller Machine) problem. Boundary Value Testing: Generalizing Boundary Value Analysis, Limitations of Boundary Value Analysis, Robustness Testing, Worst-Case Testing, Special Value Testing, Test cases for

the Triangle problem, Test cases for the NextDate function, Test cases for the Commission Problem, Random Testing and Guidelines for Boundary Value Testing.

Module -II**(12 Hours)**

Equivalence Class Testing: Equivalence Classes, Weak Normal Vs Strong Normal Equivalence Class Testing, Weak Robust Vs Strong Robust Equivalence Class Testing, Equivalence Class Test Cases for the Triangle Problem, Equivalence Class Test Cases for the Next Date Function and Equivalence Class Test Cases for the Commission Problem, Guidelines for Equivalence Class Testing. Decision Table Based Testing: Decision tables, Test cases for the triangle problem, Test cases for the Next Date function, Test cases for the commission problem, Guidelines and observations. Data flow Testing: Definition Use Testing, Example- The Commission Problem, Slice-Based Testing, Guidelines and Observations.

Module -III**(12 Hours)**

Levels of Testing: The SATM System, Structural and Behavioural Insights. Integration Testing: A Closer Look at the SATM System, Decomposition-Based Integration, Top Down Vs Bottom-Up Integration, Sandwich Integration, Call Graph-Based Integration, Pair wise Integration, Neighbourhood Integration, Path-Based Integration. System Testing: Threads, Basic concepts for requirements specification, Finding threads, Structural strategies and functional strategies for thread testing, Interaction Testing: A Taxonomy of Interactions, Static Interaction in a Single Processor, Static Interaction in Multiple Processors, Dynamic Interaction in a Single Processor, Dynamic Interaction in Multiple Processors, Client-Server Testing.

Module -IV**(12 Hours)**

Object Oriented Testing: Issues in Object Oriented Testing, Implication of Composition and Encapsulation, Implications of Inheritance, Implications of Polymorphism, GUI Testing, Object-Oriented Integration Testing. Exploratory Testing: The context-driven school, Exploring exploratory testing, Exploring a familiar example, Exploratory and context-driven testing observations. Model-Based Testing: Testing based on models, Appropriate models, Use case-based testing, Commercial tool support for model-based testing. Test-Driven Development: Test-then-code cycles, Automated test execution, Java and JUnit example, Remaining questions, Pros, cons, and open questions of TDD, Retrospective on MDD versus TDD, Software Testing Excellence: Craftsmanship, Best practice of software testing, Top 10 best practices for software testing excellence.

Self-Study topics for Advanced Learners

1. Advanced Selenium & WebDriver Techniques
2. Test Automation Frameworks: Building custom frameworks with JUnit, TestNG, or PyTest.

Textbooks

1. Paul C. Jorgensen: Software Testing, A Craftsman's Approach, 3rd Edition, 2013.

Reference books

1. Mauro Pezze, Michal Young: Software Testing and Analysis – Process, Principles and Techniques, 1st edition, John Wiley & Sons, 2011.
2. Brian Marick: The Craft of Software Testing, 1st edition, Pearson, 2012.
3. Srinivasan Desikan, Gopalaswamy Ramesh: Software testing Principles and Practices,

1st Edition, Pearson, 2012.

4. Aditya P Mathur: Foundations of Software Testing, Pearson, 2008.

3. Machine Learning [DSML61]

Module -I (12 hours)

Fundamentals of Machine Learning What is Machine Learning? Need of Machine Learning, Types of Machine Learning Systems, Main Challenges of Machine Learning, Applications of Machine Learning. Python, scikit-learn, Essential Libraries and Tools.

Module -II (12 Hours)

Data Preparation: Working with Real Data, look at the Big Picture, Get the Data, Discover and Visualize the Data to Gain Insights, Prepare the Data for Machine Learning Algorithms, Select and Train a Model.

Module -III (12 Hours)

Supervised Learning Classification and Regression, Some Sample Datasets, k-Nearest Neighbours, Linear Models, Naive Bayes Classifiers, Decision Trees. Evaluation Metrics, Ensemble learning, Bias and Variance Trade off, Handling Class Imbalance.

Module -IV (12 Hours)

Unsupervised Learning Clustering, K-Means, Limitations of K-Means, clustering for image segmentation, Clustering for Preprocessing, Clustering for Semi-Supervised Learning, DBSCAN, Other Clustering Algorithms.

Self-Study topics for Advanced Learners Clustering for image segmentation Other Clustering Algorithms

Textbooks

1. Andreas. C. Müller and S. Guido, "Introduction to Machine Learning with Python," O'Reilly, 2017
2. Amanda. Casari and Alice. Zheng, "Feature Engineering for Machine Learning," O'Reilly Media, Inc., 2018, p. 218.
3. A. Géron, "Hands-on Machine Learning with Scikit-Learn, Keras, and Tensorflow," O'Reilly Media, Inc., 2022.
- 4 Ian Good fellow, Yoshua. Bengio, and Aaron. Courville, "Deep Learning," MIT Press, 2016.

Reference books

1. S. Rashka and V. Mirdzhalili, "Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow 2," Packt, Birmingham and Mumbai, 2020.
2. S. Shalev-Shwartz and S. Ben-David, "Understanding Machine Learning: From Theory to Algorithms," Cambridge University Press, 2014.

E Books

1. Machine Learning with Python Tutorial by Bernd Klein — available as free PDF/EPUB under a Creative-Commons license. Open Tech Book+1

2. Introduction to Machine Learning with Python by Andreas C. Müller & Sarah Guido — widely used, and there are publicly accessible PDF copies online. repository.jdbikolkata.in+2codersguild.net+2

4. Mobile Application Development [DSMAD62]

Unit -I (15 Hours)

Introduction : Brief History of mobile technologies, Different mobile technologies Key Mobile Application Services-Introducing Android, The Android Application Components, Exploring the Development Environment, -Obtaining the Required Tools Launching Your First Android Application-Exploring the IDE-Debugging Your Application-Publishing Your Application Using Activities - Fragments and Intents in Android : Working with activities, Using Intents, Fragments, Using the Intent Object to Invoke Built-in Application

Unit-II (15 Hours)

Working with the User Interface Using views Understanding the Components of a Screen- Adapting to Display Orientation-Managing Changes to Screen Orientation- Utilizing the Action Bar-Creating the User Interface Programmatically Listening for UI Notification Using Basic Views-Using Picker Views -Using List Views to Display Long Lists Understanding Specialized Fragments - Using Image Views to Display Pictures -Using Menus with Views Using WebView- Saving and Loading User Preferences-Persisting Data to Files-Creating and Using Databases.

Unit-III (15 Hours)

Designing User interface Designing by declaration, creating the opening screen, using alternate resources, implementing an about box, applying a theme, adding a menu, adding settings, debugging with log messages, debugging with debugger.

Unit-IV (15 Hours)

Creating Your Own Content Providers -Using the Content Provider, SMS Messaging Sending Email-Displaying Maps- Getting Location Data- Monitoring a Location Putting SQL to work Introducing SQLite, In and Out of SQLite, Hello Database, Data Binding, using content provider, implementing content provider. Reading/writing local data, Accessing the Internal File system, Accessing the SD card. Preparing app for publishing, Deploying APK files, uploading in Market, Consuming Web Services Using HTTP- Consuming JSON Services- Creating Your Own Services Binding Activities to Services - Understanding Threading

Text Books

1. Wei-Meng Lee, Beginning android 4 application Development, John Wiley & sons, Inc, 2012
2. Jerome DiMarzio, “Beginning Android Programming with Android Studio”, 4th Edition

Reference books

1. Grant Allen, Beginning Android 4, Apress, 2012
2. Pradeep Kothari, “Android Application Development (With Kitkat Support)”, Black Book 2014.

E Books

1. https://goalkicker.com/AndroidBook/#google_vignette